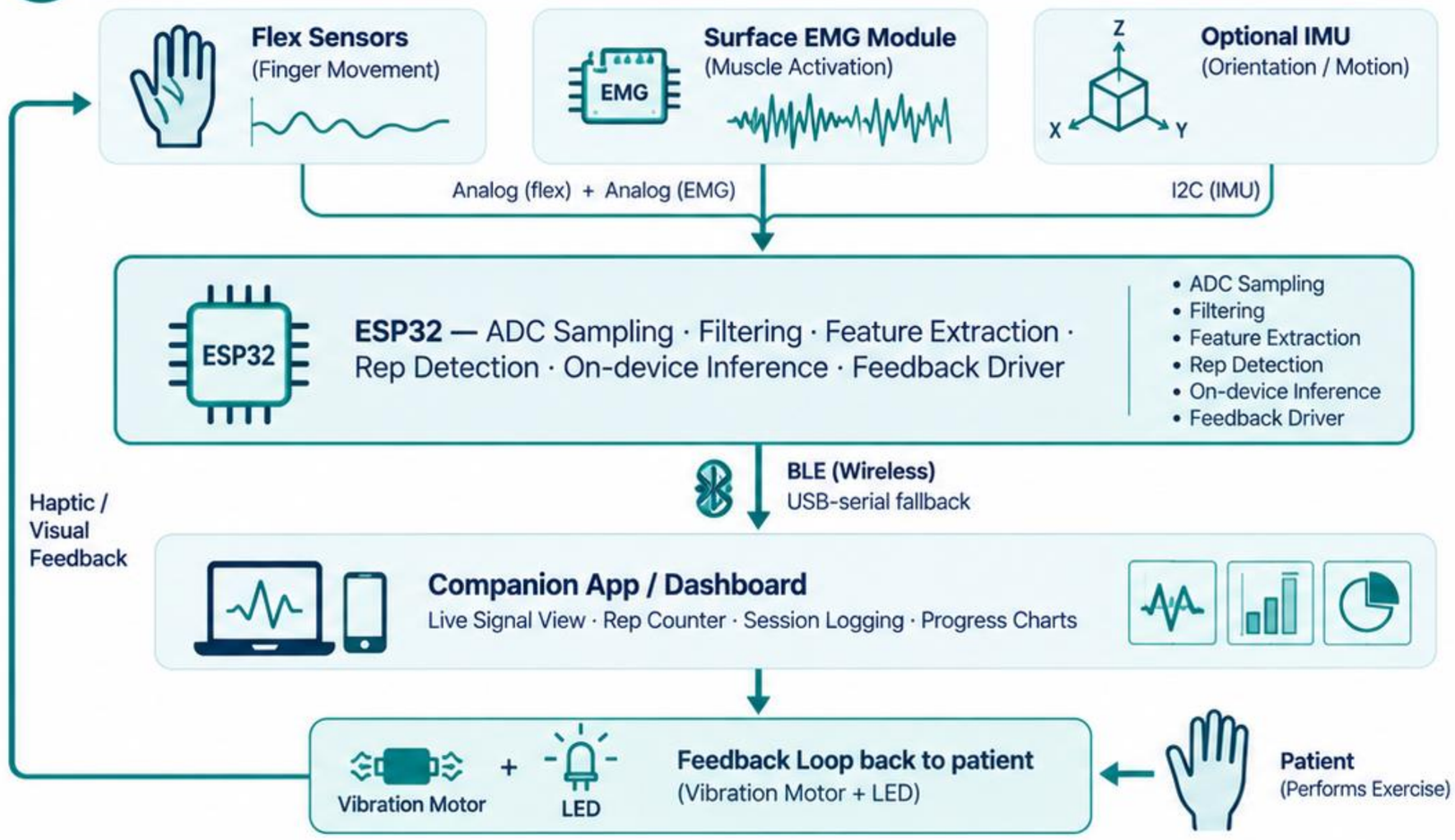
System Architecture & Technical Implementation

SENSE • PROCESS • FEEDBACK • RECOVER

From Signals to **Stronger Hands**

1 SYSTEM ARCHITECTURE



2 TECH STACK

Hardware / Embedded:
ESP32 (dual-core, WiFi/BLE), flex sensors (voltage-divider), MyoWare-class sEMG module, vibration motor + LED feedback actuators

Frontend:
Lightweight companion dashboard app — live sensor trace, rep counter, session summary, progress trend charts

Backend:
On-device signal processing (filtering, rectification, envelope extraction, feature extraction) running on ESP32; classification (Random Forest / SVM) on companion app for hackathon scope

Database:
Local session logging (compact JSON/binary per-repetition records — timestamp, ROM, EMG RMS, label)

APIs / Tools:
BLE GATT for streaming, USB-serial as wired demo fallback, scikit-learn (RF/SVM/LogReg) for classifier training